How to Build a Universe—and Find Out Whether Its Ghosts Are Real

A general-audience guide to spacetime, causality, light, clocks, and gravitational waves in conformal gravity

GPT-5.6-sol and Asger Alstrup Palm

Einstein's theory of relativity turns gravity into the geometry of spacetime. Matter and energy curve spacetime; that geometry tells matter, light and clocks how to move.

This picture explains familiar phenomena. Clocks run at different rates in different gravitational fields. Light bends and changes color. Gravitational waves cross the universe. Black holes form. The universe expands.

Our project asks whether a different mathematical theory of spacetime—called conformal or pure-Weyl gravity—can support a physically coherent universe. Its equations contain four derivatives instead of Einstein's two. That makes the theory interesting for quantum gravity, but it also produces extra solutions whose signs look wrong. These are commonly called gravitational ghosts.

We do not assume in advance that the extra solutions are physical, harmless or fatal. We build the mathematical universe in layers and make each layer pass the standard tests physicists use for spacetime, causality, waves, clocks, interactions and quantum particles.

The basic rule is:

> A pattern in an equation is not yet a physical object. It must survive the constraints, carry a measurable physical effect, propagate causally, and remain consistent when interactions and quantum physics are included.

The universe in ordinary language

The project has a mathematically complete classical starting universe for small disturbances before their full interactions are switched on.

Space is finite but has no edge. At each moment it is a three-dimensional sphere, and that whole sphere evolves through time. This is a model of an entire universe, not a box cut out of a larger space.

The closest everyday analogy is the two-dimensional surface of the Earth. It has a finite area, but a traveler can keep moving without reaching an edge. A three-dimensional sphere is the same closed idea one dimension higher.

Stacking the successive spherical spaces through time produces what mathematicians call the conformal cylinder. The word *cylinder* describes the history of a closed universe. It does not mean that space is shaped like a pipe or that the universe sits inside one.

The short version of what we have

- **Cause runs from past to future.** If a source is switched on tomorrow, the tested classical equations do not let it change a detector today. This has been checked for the complete gravitational system and for the version containing the matter clock—not merely for one convenient equation.
- **There are physical clocks.** Healthy matter fields provide an internal reading that changes steadily. In a particular zero-total-charge version of the closed universe, shifting the time label of the entire history can be a redundant description even while the internal clock keeps changing. In the unrestricted version, the same time shift carries a real charge and is a physical symmetry.
- **There are classical light and gravitational-wave directions.** In a coupled gravity-and-electromagnetism universe, familiar electromagnetic and gravitational wave patterns solve the linear

equations and carry a nonzero physical comparison rule. This establishes classical waves, not quantum photons, quantum gravitons or a detector prediction.

- **The larger theory also has two extra classical wave directions in one tested family.** They are not disguised Einstein waves, and an exact local comparison between them is nonzero and positive at the current reduced equation level. This means they do not vanish as mere algebraic bookkeeping there. It does **not** yet mean that nature contains two new particles: the full spacetime energy rule, causal boundary conditions and final physical quotient still have to be checked.
- **The apparent ghost is not automatically a particle.** In the selected boundary-free, zero-charge universe, no isolated one-particle conformal graviton survives after redundant descriptions are removed. Two collective curvature patterns remain with a positive comparison rule. They describe possible interactions or changes to the theory, not particles flying through space.
- **The first interaction test works in the clock universe.** The quadratic interaction satisfies the required gauge and consistency identities. This is the first nonlinear rung, not a proof that every higher interaction is stable.
- **Every statement has a computational receipt.** Exact symbolic programs derive and check the large identities. Separate verifiers, broken-input tests, content hashes and clean rebuilds record what was proved and prevent a local calculation from being advertised as a quantum or cosmological result.

This is a real but incomplete mathematical universe. It has spacetime, classical causal propagation, clocks, classical electromagnetic and gravitational wave directions, two classified extra fourth-order directions in a separate compact gravity-and-electromagnetism setting, and a first controlled interaction layer. It does not yet have a certified relational redshift, electrons, quantum particles, gravitational lensing, black-hole boundaries, a scattering experiment, or a dark-matter or dark-energy prediction.

What “gauge” means

Gauge is the physicists’ word for redundancy in a description. It means that two different-looking sets of mathematical variables represent the same physical situation.

The same place on Earth can be described using latitude and longitude, a street map or a rotated grid. The numbers change, but the place does not. Gravity has the same feature: stretching or relabeling the coordinate grid can change every component of the metric without changing the underlying spacetime. Conformal gravity has an additional freedom that changes the local scale—the choice of ruler—while preserving the light cones.

A gauge transformation is not a new force, event or wave. Counting it as physical would count the same universe more than once. Removing gauge means identifying all descriptions that differ only by this redundancy.

There is an important difference between a gauge transformation and a physical symmetry. They can look similar in the equations. A physical symmetry has a measurable generator or charge, such as energy or angular momentum. A proper gauge direction has zero generator in the declared physical setting and changes no observable. The calculation decides which case applies; the label is not chosen by taste.

Asking whether time translation is gauge does **not** mean that clocks stop, change is unreal or every moment is identical. It asks whether shifting the coordinate label of the entire closed universe creates a new physical state, or only a new description of the same constrained history. Relational statements—what one field does when a physical clock reads a particular value—can remain nontrivial even when the total shift is gauge.

This distinction changes the ghost question. A raw extra solution can be a genuine physical direction, a redundant description, a mode removed by a constraint, or a boundary excitation. Its sign matters physically only after that classification is complete.

What causality means here

A causal theory does not allow an event in the future to change what already happened in the past. If a source is switched on tomorrow, a detector must not respond today. A disturbance may influence only events inside its future light cone.

The physical response used for this test is the **retarded** response: it starts at the source and propagates toward the future. Physicists also construct an **advanced** mathematical partner pointing toward the past. That partner is needed to test the equations and build physical comparison rules; it does not authorize backward-in-time signalling. The physical source-response rule is retarded.

The complete free pure-Weyl calculation passes this classical causality test across all 386 linked parts of its gravitational bookkeeping system. The clock-coupled Berger universe has a separate causal result across all 54 parts of its gauge-fixed system. In both cases the retarded response is confined to the causal future and the constraints propagate with it.

These are classical causality results. A global quantum state with the required short-distance and causality properties is a separate rung.

Work toward that quantum rung has identified the standard local short-distance wave pattern for the basic clock-universe fields and derived an exact formal recipe for transporting it through the coupled equations. The recipe is not yet a quantum state. The current mathematical estimates do not prove that all singular directions of the transported distributions fit together safely. The certification therefore stops at that precise boundary.

What each rung has to prove

Words such as *light*, *particle* and *black hole* carry a great deal of physics. We do not count a phenomenon merely because something in an equation has a familiar shape. Each rung has a recognized acceptance test.

Spacetime and curvature

The geometry must solve the gravitational field equations. Small disturbances must obey compatible equations and constraints. This is what gives defined meaning to distances, durations, curvature and light cones.

Causality

Sources must produce responses only where the light cones permit them. In particular, a future source must not alter the past. For gravity this must hold for the entire constrained gauge system, not just one selected component.

Light

Classical light requires electromagnetic equations, nontrivial wave solutions and propagation along the spacetime light cones. Observable light also requires sources, detectors and energy flux. A photon requires the additional quantum-particle tests below.

Electrons and charged matter

An electron-like field must have the correct spin, electric charge and fermionic behavior. It needs causal propagation, stable coupling to electromagnetism and gravity, and a physical mechanism that supplies a mass scale in a conformal theory. A classical spinor field is still not a quantum electron until the particle tests pass.

Gravitational waves

A gravitational wave must solve the linearized gravitational equations, survive all redundant coordinate and scale descriptions, carry a nonzero physical comparison rule, and propagate causally. An observable waveform also needs energy flux, boundary conditions and a response in a detector.

Clocks, time dilation and redshift

A clock must change steadily, have healthy energy and couple consistently to gravity. Time dilation requires comparing two such physical clocks. Gravitational redshift requires an emitter, a light signal and a receiver; the observable is the ratio of the frequencies measured by the two clocks, stated without relying on arbitrary coordinate labels.

Gravitational lensing

Lensing requires a physical source of curvature, causal light paths around that source and observable comparisons of angles, brightness or arrival times. Drawing a bent line in a coordinate system is not enough—the result must be independent of how the spacetime map was drawn.

Interactions

The nonlinear equations must say how waves and matter exchange energy. The interactions must preserve all constraints and must not regenerate a forbidden or unstable physical mode. Passing quadratic order does not automatically grant cubic order, higher orders or global evolution.

Quantum particles

A particle requires a global quantum state, a positive probability rule and a controlled separation into physical and redundant states. Scattering particles additionally require meaningful incoming and outgoing regions. A classical wave is therefore evidence for a field mode, not yet for a photon, electron or graviton.

Black holes

A black-hole candidate must possess a genuine horizon rather than a coordinate artifact. Its boundary conditions, conserved charges, causal perturbations, thermodynamic quantities and stability must all be controlled. A metric with a zero in one coefficient is not by itself a physical black hole.

Quantum gravity

The quantum gauge identities must remain consistent after regularization and renormalization. Quantum anomalies must either cancel or have an allowed repair. The theory also needs suitable short-distance states, causal quantum products and a physical state space. Classical causal propagation is an essential input, not the completed quantum theory.

Cosmology, dark matter and dark energy

The theory must supply stable evolving universes and galaxy-scale solutions, derive observables such as expansion, lensing and rotation curves, and compare them with data. Fitting a curve is meaningful only after the underlying modes, clocks, causality and stability have passed their own tests.

The universe-building map

The table below is both a status summary and a public TODO list. A scoped pass means that the stated phenomenon exists in a precisely declared mathematical setting. It does not automatically extend to every background, interaction or quantum theory.

Familiar feature	Where the project stands	Decisive next test
Spacetime and curvature	**Scoped pass.** Exact boundary-free spherical and Berger backgrounds solve their declared classical equations.	Extend causal control to broader globally hyperbolic backgrounds.
Causality	**Scoped pass.** Retarded responses in the complete 386-part gravity system and 54-part clock system do not let future sources alter the past.	Construct the corresponding global quantum state and quantum causality theorem.
Clocks and time dilation	**Partial.** A healthy matter clock changes internally while total time shift can remain gauge in the fixed-coupling, linear, zero-charge sector.	Compare two physical clocks and calculate an observable time-dilation law.

Familiar feature	Where the project stands	Decisive next test
Gravitational redshift	**Open, with ingredients present.** The model has curvature, causal propagation, light-like paths and a matter clock.	Connect emitter, light signal and receiver into one gauge-invariant frequency ratio.
Classical light	**Partial.** Standard electromagnetic wave directions occur in the compact Einstein–Maxwell inclusion.	Build physical sources, detectors, energy flux and boundary conditions.
Electrons and charged matter	**Open.** No certified charged spin-one-half matter sector exists in the current universe.	Add a Dirac field, a physical mass/scale mechanism, causal propagation and stable interactions.
Gravitational waves	**Partial.** Standard linear gravitational wave directions occur with a nonzero physical pairing, and the pure-gravity complex propagates causally. In a separate compact axial family, two extra fourth-order directions survive the reduced equation and local-pairing tests.	Match the extra directions to the direct four-dimensional physical current, impose causal boundary conditions, and produce measurable waveforms, flux and detector response.
Gravitational lensing	**Open, with geometric ingredients present.** Curved spacetime and light cones exist, but no certified lensing observable does.	Add a localized lens, propagate light around it and compare observable angles and arrival times.
Quantum particles	**Open.** Classical waves are not yet photons, gravitons or electrons; the surviving curvature classes are not particles.	Construct a global quantum state, physical positive pairing and incoming/outgoing particle interpretation.
Interactions	**Partial.** The complete quadratic clock-coupled interaction passes its declared cyclic and gauge tests.	Pass cubic and higher identities, resonant channels and global nonlinear evolution.
Black holes	**Open in this certified pipeline.** No horizon phase space, boundary charge or stability theorem has been imported into the universe.	Select a black-hole background and certify its horizons, causal perturbations, charges, entropy and stability.
Quantum gravity	**Early groundwork only.** Candidate anomaly types and the basic local short-distance wave structure are partly classified. Exact formal transport through the clock system is known, but its distributional wavefront safety is not; there is no coupled Hadamard state or restored quantum master equation.	Prove or obstruct the microlocal transport, then compute anomaly coefficients, restore the quantum gauge identity and construct the global quantum theory.
Cosmology, dark matter and dark energy	**Open.** The current work establishes consistency machinery, not a fitted cosmological model.	Build stable cosmological and galaxy backgrounds, derive observables, then compare them with data.

If the universe differs from standard physics

Agreement with Einstein gravity and ordinary quantum field theory is one possible result. It would show that the new mathematical framework can recover known physics while organizing its gauge and causal structure differently.

A controlled difference could be new physics. An extra wave, charge, interaction or cosmological effect would be interesting only if it passes the same causality, stability, gauge and quantum tests as the familiar phenomena.

The two newly isolated extra wave directions are now at exactly this fork. They have survived enough exact tests that they cannot be dismissed as a repeated factor in an equation, but not enough to call them observable new physics. The next physical comparison may show that they are healthy, ghostly, removed by admissible boundaries, or confined to this compact sector.

A difference can also be a no-go result. The calculation may show that a clock cannot remain healthy, a mode has an unavoidable negative physical direction, an interaction cannot preserve the constraints, a black-hole boundary condition is inconsistent, or a quantum anomaly cannot be removed.

Sector dependence is another possible outcome. The theory may work on a closed zero-charge universe but fail when time translation carries energy at an outer boundary. That is not a contradiction; it is a statement that the physical theory requires a selection rule and does not describe every possible universe in the same way.

The project can therefore stop at any rung. If it stops because an exact obstruction is found, that obstruction is still valuable physics. It tells us which attractive idea cannot describe nature, under which assumptions, and what a successful theory would have to change. If it continues, every passed rung narrows the space of viable alternatives and makes a possible new prediction more credible.

Success and failure are both scientific outcomes:

- **recovery** shows that known physics is contained;
- **new physics** is a difference that survives all required tests;
- **sector selection** identifies the conditions under which the theory works;
- **a scoped no-go** rules out a complete family of proposed constructions;
- **a decisive obstruction** marks the rung where this candidate universe ends.

The code is a symbolic physics laboratory

The calculations are too large and too sensitive to signs and conventions to manage reliably as hand algebra alone. We use code to perform exact symbolic derivations while keeping the physical assumptions visible.

This is not a numerical simulation of a universe. The programs derive and check algebraic consequences of the equations. They generate differential operators from the action, expand large interactions coefficient by coefficient, and compute ranks, constraints, pairings, signs and obstructions using exact rational or algebraic arithmetic.

The scale is substantial. The free causal gravity model links 386 rows of fields, equations and gauge data. The clock-coupled interaction uses a 54-part system and contains 54,236 canonical nonzero coefficients after exact simplification.

Every material claim carries an accountability trail:

1. The theory, background, charge sector, boundary conditions and claim level are declared.
2. A machine-readable certificate records inputs, assumptions, hashes, outputs and unresolved fields.
3. An independent verifier replays the decisive identity instead of trusting the program that generated it.
4. Deliberately broken inputs must fail, preventing a verifier from passing without actually checking the mathematics.
5. Local algebra, causal physics and quantum claims receive different labels and cannot be silently exchanged.
6. Publication files are rebuilt from a clean archived commit and compared with recorded hashes.
7. Failed approaches and no-go results remain in the ledger instead of disappearing when another route succeeds.

The repository therefore functions like a symbolic experimental laboratory. A result can be inspected at three levels: the physical statement, the mathematical identity and the exact computational receipt.

The code cannot prove assumptions that were never encoded. Analytic existence, global boundary conditions and quantum interpretation must each pass their own tests. The system is designed to fail closed when a required layer is missing.

Where the technical continuation lives

This article is the general-audience front door. The specialist continuation is maintained separately so that this explanation does not gradually turn into a technical paper.

The physicist executive summary gives the claim-by-claim result spine, audience-specific connections, lifecycle labels and direct links to the papers and certificates. A PDF version is available as a short technical briefing.

The central research question is:

> Is the apparent extra gravitational branch a physical instability, a gauge direction, a constrained sector, a boundary excitation or a quantum anomaly—and which answer applies in each mathematically complete universe?

The project builds enough of each universe to make that question testable. Passing a rung expands the universe. Failing a rung produces an exact, reproducible boundary on what the theory can describe.